Tubeless tyre

This tyre does not have any tube inside it. In this type of tyre, the tyre is directly fitted on the rim. The cross sectional view of the tubeless tyre can be seen in the Figure.

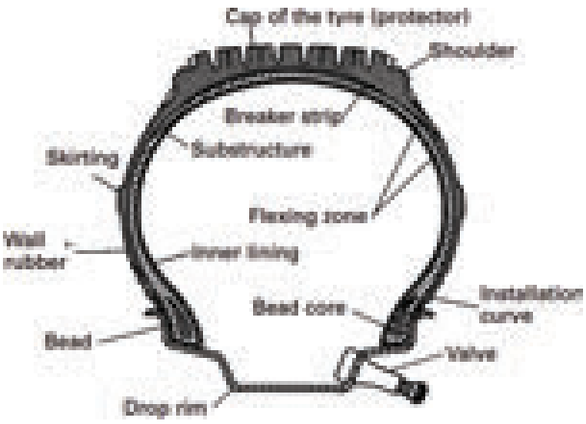


Figure Construction of a tubeless tyre

Advantages

1. Simple in structure.
2. In case of puncture, the tyre can be repaired without removing from the rim.
3. The tyre needs no tube.
4. Maintenance is low.

Bullet proof tyre

In this type of tyres even if the tyre is punctured by the bullet, the air pressure inside the tyre does not decrease and the puncture is automatically corrected by the tyre itself. This type of tyre is used in military vehicles.



Figure Photographic view of the bullet proof tyre

Liquid filled tyre

In this type of tyre the mixture of 15% of calcium chloride 85% of water is used to fill the tyre for about 75% of the total capacity of the tyre. This type of tyre is used in some heavy vehicles and tractors.

5.6 Properties of tyre

The following are the important properties of the tyres

1. It should be more comfort for traveling.
2. It should be able to withstand vibrations of the road sides.
3. The tyre should give stability to the vehicle when the vehicle is negotiating a turn.
4. Vehicle should stop immediately by creating the friction without any slip when applying the brake.
5. It should be able to give very less rolling resistance.
6. The tyre must provide safety to the vehicle when the vehicle moving on the road.
7. The tyre must work for a long period.

5.7 Functions of the tyres

1. It withstands the whole weight of the vehicle.
2. It creates the friction on the road and helps to roll the wheel on the road.
3. It gives road stability for the vehicles.

BAE_ENG_Chapter_05.indd 79 12/20/2019 1:02:44 PM